<u>SudoKoders</u>
(Sudoku Board Generator)

SAMPLE INPUT AND OUTPUT

CASE 1:Valid Move

```
C:\Users\diyan\Documents\SudoKoders>python "Sudoku Generator".py
Generated puzzle:
[0, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
Type -1 in row to quit and -2 in row to show the full solved puzzle.

To change the value which you added ,go to the specific cell and then give the desired value or set it to zero to erase the value.


Enter your move :
Row (0-8): 0
Column (0-8): 0
Number (1-9): 4

Valid move!

Updated Puzzle:
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]

Enter your move :
```

CASE 2:Invalid Move

```
Updated Puzzle:
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]

Enter your move :
Row (0-8): 0
Column (0-8): 1
Number (1-9): 5

Invalid move! Puzzle unchanged. Try again...
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
Row (0-8): 0
Column (0-8): 0
Number (1-9): 0
```

<u>Case 3</u>:Erasing a previously entered value

```
[4, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
Row (0-8): 0
Column (0-8): 0
Number (1-9): 0

Valid move!

Updated Puzzle:
[0, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]

Enter your move :
```

<u>Case 4</u>:Replacing a previously entered value

```
Updated Puzzle:
[0, 0, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]

Enter your move :
Row (0-8): 0
Column (0-8): 1
Number (1-9): 1

Valid move!

Updated Puzzle:
```

```
[0, 1, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]
Row (0-8): 0
Column (0-8): 1
Number (1-9): 4

Valid move!

Updated Puzzle:
[0, 4, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]

Enter your move :
```

<u>Case 5:</u>Generating a complete solution of the puzzle

```
Updated Puzzle:
[0, 4, 0, 0, 0, 0, 3, 7, 0]
[2, 0, 0, 4, 0, 0, 0, 5, 1]
[0, 7, 0, 3, 5, 9, 8, 0, 0]
[0, 0, 7, 5, 1, 2, 4, 0, 0]
[0, 2, 0, 6, 0, 7, 5, 1, 0]
[0, 5, 0, 0, 0, 0, 0, 2, 0]
[0, 8, 0, 7, 6, 1, 0, 9, 0]
[0, 9, 0, 2, 0, 4, 1, 6, 0]
[6, 0, 2, 8, 9, 5, 7, 3, 4]

Enter your move :
Row (0-8): -2
Solved Sudoku:
[5, 4, 9, 1, 2, 8, 3, 7, 6]
[2, 3, 8, 4, 7, 6, 9, 5, 1]
[1, 7, 6, 3, 5, 9, 8, 4, 2]
[9, 6, 7, 5, 1, 2, 4, 8, 3]
[4, 2, 3, 6, 8, 7, 5, 1, 9]
[8, 5, 1, 9, 4, 3, 6, 2, 7]
[3, 8, 4, 7, 6, 1, 2, 9, 5]
[7, 9, 5, 2, 3, 4, 1, 6, 8]
[6, 1, 2, 8, 9, 5, 7, 3, 4]
Enter your next move:

Enter choice 1 to generate a new puzzle:

Enter choice -1 to quit

To change the value which you added ,Change it to zero and then give the desired value.
Enter choice: 1
New Puzzle Generated:
[0, 0, 2, 8, 0, 0, 0, 0, 0]
[0, 0, 1, 2, 5, 3, 6, 4, 8]
[0, 3, 4, 0, 0, 0, 0, 0, 0]
[2, 1, 9, 0, 8, 5, 0, 3, 0]
[0, 0, 0, 0, 1, 0, 4, 0, 5]
[4, 6, 0, 9, 0, 7, 0, 8, 1]
[9, 0, 6, 5, 0, 8, 0, 0, 3]
[0, 8, 7, 0, 0, 1, 9, 6, 0]
[0, 0, 0, 7, 6, 9, 0, 5, 4]
```

<u>Case 6:</u>Quitting the game

```
New Puzzle Generated:
[0, 0, 2, 8, 0, 0, 0, 0, 0]
[0, 0, 1, 2, 5, 3, 6, 4, 8]
[0, 3, 4, 0, 0, 0, 0, 0, 0]
[2, 1, 9, 0, 8, 5, 0, 3, 0]
[0, 0, 0, 0, 1, 0, 4, 0, 5]
[4, 6, 0, 9, 0, 7, 0, 8, 1]
[9, 0, 6, 5, 0, 8, 0, 0, 3]
[0, 8, 7, 0, 0, 1, 9, 6, 0]
[0, 0, 0, 7, 6, 9, 0, 5, 4]
Row (0-8): 0
Column (0-8): 0
Number (1-9): 5

Valid move!

Updated Puzzle:
[5, 0, 2, 8, 0, 0, 0, 0, 0]
[0, 0, 1, 2, 5, 3, 6, 4, 8]
[0, 3, 4, 0, 0, 0, 0, 0, 0]
[2, 1, 9, 0, 8, 5, 0, 3, 0]
[0, 0, 0, 0, 1, 0, 4, 0, 5]
[4, 6, 0, 9, 0, 7, 0, 8, 1]
[9, 0, 6, 5, 0, 8, 0, 0, 3]
[0, 8, 7, 0, 0, 1, 9, 6, 0]
[0, 0, 0, 7, 6, 9, 0, 5, 4]

Enter your move :
Row (0-8): -1
You quit the game!
```